

Abstract & Introduction

Retrieval-Augmented Generation (RAG) is a hybrid NLP architecture that combines parametric memory (a pre-trained language model) with non-parametric memory (a dense retrieval index over a document corpus). RAG models retrieve relevant documents at inference time and condition generation on those documents, achieving strong performance on knowledge-intensive tasks without requiring the model to memorise all world knowledge during pre-training.

Traditional closed-book generation models such as GPT-2 and T5 rely entirely on knowledge encoded in model weights. This approach leads to hallucinations when the required knowledge was absent or outdated in the training data. RAG addresses this by grounding generation in retrieved evidence, dramatically reducing factual errors and enabling real-time knowledge updates by swapping the document index.

This paper provides an overview of the RAG paradigm, its core components, evaluation methodology, and key findings across several benchmark datasets including Natural Questions, TriviaQA, WebQ, and CuratedTrec.

Related Work

Dense Passage Retrieval (DPR) introduced bi-encoder retrieval using dual BERT encoders trained on question-passage pairs. DPR showed that learned dense representations outperform classical sparse methods such as BM25 on open-domain QA benchmarks.

REALM (Retrieval-Enhanced Language Model) pre-trains a language model jointly with a retriever in an end-to-end masked language modelling objective, demonstrating that retrieval can be integrated into pre-training. However, REALM is computationally expensive.

FiD (Fusion-in-Decoder) extends T5 with multi-document fusion, encoding each retrieved passage independently and fusing them in the decoder cross-attention. FiD achieves state-of-the-art results on NaturalQuestions and TriviaQA.

Unlike these, RAG integrates retrieval directly into the generation process using a trainable retriever and a seq2seq generator, providing end-to-end differentiability.

Architecture & Methodology

The RAG model has two main components:

1. Retriever: A DPR bi-encoder that maps queries and documents to a shared embedding space. Given a query q , the retriever returns the top-k documents $z_1, \dots, z_k$ from a pre-built FAISS index over a large document corpus (e.g., Wikipedia).
2. Generator: A BART-large seq2seq model that conditions generation on the concatenation of the query and each retrieved document. Two variants are proposed:
 - RAG-Sequence: marginalises over documents for the entire sequence.
 - RAG-Token: marginalises over documents for each generated token.

The index is built offline using FAISS IndexFlatIP over 100-word document chunks from the December 2018 Wikipedia dump, resulting in 21 million passages. Retrieval is performed in sub-second latency with an approximate MIPS index.

During training, the retriever and generator are jointly optimised by minimising the negative log-likelihood of the target answer, marginalised over the top-k retrieved documents. The FAISS index is periodically refreshed to reflect updated document embeddings.

Experiments & Results

RAG models are evaluated on four open-domain QA benchmarks:

NaturalQuestions (NQ): RAG-Token achieves 44.5 EM, outperforming the closed-book T5-1.1B (36.6 EM) and the DPR reader (41.5 EM).

TriviaQA: RAG-Token achieves 56.8 EM, close to the state-of-the-art at time of publication and substantially above the T5-1.1B baseline.

WebQ: RAG-Token achieves 45.5 EM, a strong result given that WebQ contains questions with structured Freebase answers.

CuratedTrec: RAG-Sequence achieves 68.0 EM, benefiting from the sequence-level margin maximisation which suits fact retrieval tasks.

On the MSMARCO passage ranking task, RAG-Sequence achieves state-of-the-art results in the abstractive generation category, demonstrating that it can generate fluent natural language answers rather than merely extracting spans.

Human evaluation shows RAG answers are rated as more factual and more specific than closed-book T5 answers, confirming that grounding generation in retrieved evidence leads to higher quality outputs.

Limitations & Discussion

While RAG demonstrates strong empirical results, several limitations remain:

Knowledge Staleness: The document index is static. For rapidly changing domains (e.g. , recent events), the index must be rebuilt frequently, which is computationally expensive at Wikipedia scale.

Retrieval Failures: When the correct passage is not retrieved in the top-k, the generator has no evidence to ground its answer, leading to hallucination. Increasing k improves recall but increases latency.

Multi-hop Reasoning: RAG struggles on tasks requiring reasoning across multiple documents simultaneously, as each retrieved passage is encoded independently by BART.

Domain Shift: Performance degrades when the query distribution differs significantly from the Wikipedia-based index. Custom domain corpora require rebuilding the index and potentially fine-tuning the retriever.

Interpretability: The marginalisation over top-k documents makes it difficult to attribute individual generation decisions to specific retrieved passages.

Future work includes dynamic index updates, improved multi-hop reasoning via iterative retrieval, and uncertainty-aware generation that signals low-confidence answers.

Conclusion & References

We presented Retrieval-Augmented Generation (RAG), a general-purpose architecture for knowledge-intensive NLP. RAG combines a learned dense retriever with a seq2seq generator, jointly trained end-to-end. Experiments show that RAG achieves state-of-the-art results on multiple open-domain QA benchmarks, outperforming both large closed-book models and extractive readers.

The key insight is that language models do not need to memorise all world knowledge; instead, they can reliably ground generation in retrieved evidence, leading to more accurate, specific, and verifiable answers.

References

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